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2021

Q-No. 01

-i) Mean Deviation of normal Distribution

$$M.D = |x - \mu|$$

$$\begin{aligned} f(x) &= |x - \mu| \cdot f(x) dx \\ &= \int_{-\infty}^{\infty} |x - \mu| \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} |x - \mu| e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \end{aligned}$$

$$\text{let } \frac{x - \mu}{\sigma} = z \rightarrow x - \mu = \sigma z$$

$$x = \mu + \sigma z \rightarrow dx = \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \sigma z e^{-\frac{1}{2}z^2} \sigma dz$$

$$= \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{1}{2}z^2} dz$$

$$\text{let } \frac{z^2}{2} = t$$

$$z^2 = 2t \rightarrow z dz = dt$$

$$dz = dt / \sqrt{2t}$$

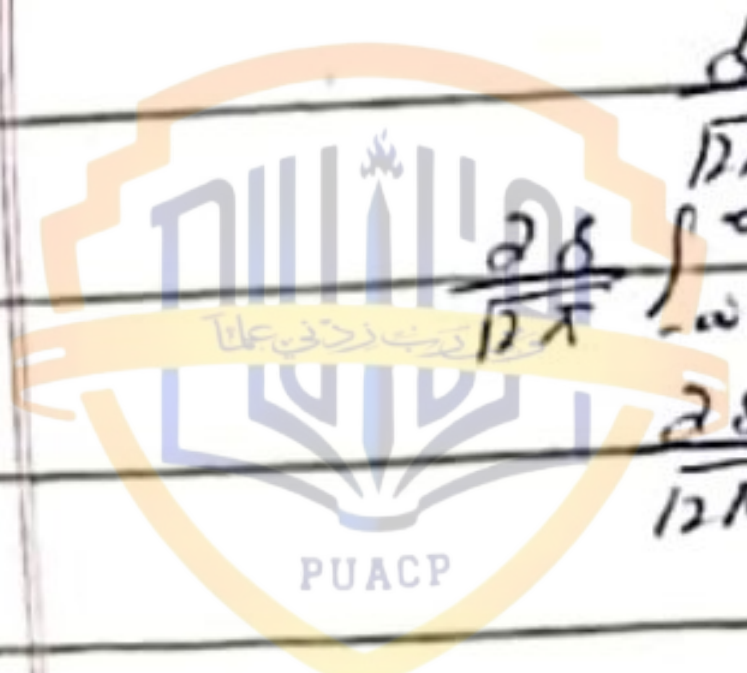
$$\frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{2t} e^{-t} \frac{dt}{\sqrt{2t}}$$

$$\frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t} dt \quad \text{even function}$$

$$\frac{2\sigma}{\sqrt{2\pi}} \int_0^{\infty} e^{-t} t^{1-1} dt$$

$$\frac{2\sigma}{\sqrt{2\pi}} \Gamma$$

$$\frac{\sqrt{2}\sigma}{\sqrt{\pi}} \Rightarrow \frac{4}{\sqrt{\pi}} \sigma \text{ M.D}$$



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-iii) r^{th} moment of Weibull Distribution

$$f(x) = abx^{b-1} e^{-ax^b}$$

$$\mu_r' = \int_0^\infty x^r f(x) dx$$

$$= \int_0^\infty x^r abx^{b-1} e^{-ax^b} dx$$

$$= ab \int_0^\infty x^{r+b-1} e^{-ax^b} dx$$

$$x = (z/a)^{1/b} \quad \Rightarrow \quad z = ax^b \quad \Rightarrow \quad x = (z/a)^{1/b}$$

$$\text{let } ax^b = z \Rightarrow x = (z/a)^{1/b}$$

$$= ab \int_0^\infty \left[(z/a)^{1/b} \right]^{r+b-1} e^{-z} \frac{1}{ab} (z/a)^{1/b-1} dz$$

$$= \int_0^\infty (z/a)^{r/b} \cdot (z/a)^{1/b-1} e^{-z} (z/a)^{1/b-1} dz$$

$$= \int_0^\infty (z/a)^{r/a} e^{-z} dz$$

$$= (1/a)^{r/a} \int_0^\infty z^{(r/a)+1-1} e^{-z} dz$$

$$\mu_r' = (1/a)^{r/a} \Gamma(r/a + 1)$$

-iv Mean variance of beta Distribution

$$f(x) = \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

Mean

$$= \int_0^1 x \cdot \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} dx$$

$$= \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^{(\alpha+1)-1} (1-x)^{\beta-1} dx$$

$$= \frac{1}{\beta(\alpha, \beta)} \beta(1+\alpha, \beta)$$

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$$\frac{1}{\beta(\alpha, \beta)} \quad \beta(1+\alpha, \beta)$$

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \quad \frac{\Gamma(1+\alpha+\beta)}{\Gamma(1+\alpha)\Gamma(\beta)}$$

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)} \quad \frac{\alpha(1+\alpha)\Gamma(\alpha)}{(\Gamma(\alpha+\beta))(\alpha+\beta)\Gamma(\alpha+\beta)}$$

$$\text{Mean} = \frac{\alpha(1+\alpha)}{(\alpha+\beta)(\alpha+\beta)} \quad \frac{\alpha}{\alpha+\beta}$$

Variance

$$f(x) = \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

$$= \int_0^1 x^2 \cdot f(x) \cdot dx$$

$$= \int_0^1 x^2 \frac{1}{\beta(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$$

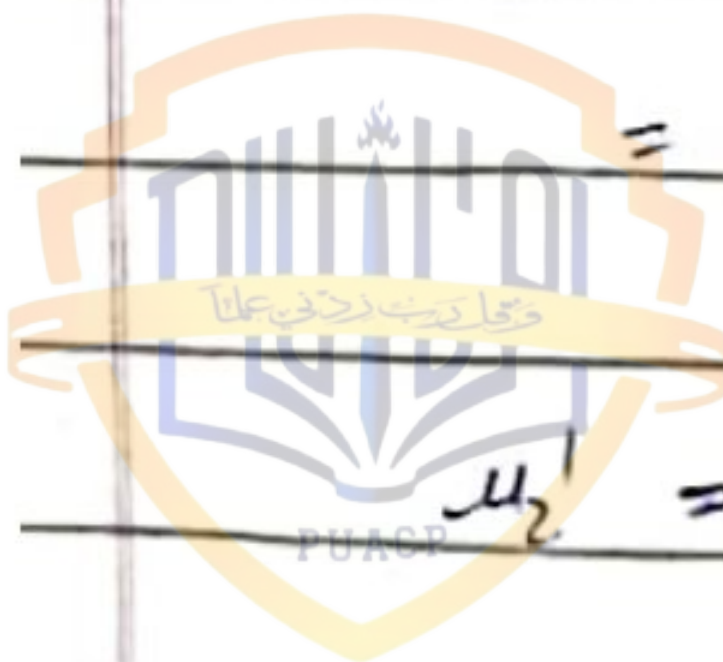
$$= \frac{1}{\beta(\alpha, \beta)} \int_0^1 x^{(\alpha+2)-1} (1-x)^{\beta-1}$$

$$= \frac{1}{\beta(\alpha, \beta)} \cdot \beta(\alpha+2, \beta)$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+2+\beta)}$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)} \frac{(\alpha+1)\alpha\Gamma(\alpha)}{(\alpha+2+\beta)(\alpha+\beta)\Gamma(\alpha+\beta)}$$

$$\mu_2 = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+1+\beta)}$$



$$\text{Variance} = \mu_2' - (\mu_1')^2$$

$$= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2$$

$$= \frac{\alpha}{\alpha+\beta} \left[\frac{\alpha+1}{\alpha+\beta+1} - \frac{\alpha}{\alpha+\beta} \right]$$

$$= \frac{\alpha}{\alpha+\beta} \left[\frac{(\alpha+1)(\alpha+\beta) - \alpha(\alpha+\beta+1)}{(\alpha+\beta+1)(\alpha+\beta)} \right]$$

$$= \frac{\alpha}{\alpha+\beta} \left[\frac{\alpha^2 + \alpha\beta + \alpha + \beta - \alpha^2 - \alpha\beta - \alpha}{(\alpha+\beta+1)(\alpha+\beta)} \right]$$



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$$\text{Variance} = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

v) m.g.f of chi-square Distribution

$$M_0(t) = \int_0^\infty e^{tx^2} f(x^2) dx^2$$

$$f(x^2) = \frac{1}{2^{n/2} \Gamma(n/2)} e^{-x^2/2} (x^2)^{n/2-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^\infty e^{tx^2} \cdot e^{-x^2/2} (x^2)^{n/2-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^\infty e^{-x^2(1/2 - t)} (x^2)^{n/2-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^\infty e^{-\frac{1}{2} x^2 [1/2 - 1]} (x^2)^{n/2-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^\infty e^{-\frac{1}{2} x^2 [1/2 - 1]} (x^2)^{n/2-1} dx^2$$

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^\infty e^{-\frac{1}{2} x^2 [1/2 - 1]} (x^2)^{n/2-1} dx^2$$

$$x^2 (1 - 2t) = z$$

$$x^2 = \frac{z}{1-2t}$$

$$\frac{1}{2} x^2 (1 - 2t) = t$$

$$x^2 = \frac{2t}{1-2t}$$

$$dx^2 = \frac{2dt}{1-2t}$$

Now

$$= \frac{1}{2^{n/2} \Gamma(n/2)} \int_0^\infty e^{-t} \left(\frac{2t}{1-2t} \right)^{n/2-1} \left(\frac{2}{1-2t} \right) dt$$

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$$\frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^1 e^{-t} \left(\frac{2t}{1-2t} \right)^{\frac{n}{2}-1} \left(\frac{2}{1-2t} \right) dt$$

$$\frac{2^{\frac{n}{2}-1} \Gamma(\frac{n}{2})}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^1 e^{-t} t^{\frac{n}{2}-1} dt$$

$$\frac{(1-2t)^{-\frac{n}{2}}}{\Gamma(\frac{n}{2})} \int_0^1 e^{-t} t^{\frac{n}{2}-1} dt$$

$$\frac{(1-2t)^{-\frac{n}{2}}}{\Gamma(\frac{n}{2})}$$

$$M_0(-1) = (1-2t)^{-\frac{n}{2}}$$

vi) Median of Cauchy Distribution

$$f(x) = \frac{\lambda}{\pi [\lambda^2 + (x-\mu)^2]}$$

$$\int_{-\infty}^{\infty} \frac{\lambda}{\pi [\lambda^2 + (x-\mu)^2]} dx = \frac{1}{2}$$

$$\frac{\lambda}{\pi} \int_{-\infty}^{\infty} \frac{1}{\lambda^2 + (x-\mu)^2} dx = \frac{1}{2}$$

$$\frac{\lambda}{\pi} \left[\frac{1}{\lambda} \tan^{-1} \frac{x-\mu}{\lambda} \right]_{-\infty}^{\infty} = \frac{1}{2}$$

$$\frac{1}{\pi} \left[\tan^{-1} \frac{\infty-\mu}{\lambda} - \tan^{-1} \frac{-\infty-\mu}{\lambda} \right] = \frac{1}{2}$$

$$\frac{1}{\pi} \left[\tan^{-1} \frac{\infty-\mu}{\lambda} - \tan^{-1} \frac{-\infty-\mu}{\lambda} \right] = \frac{1}{2}$$

$$\tan^{-1} \left(\frac{m - \mu}{\lambda} \right) = 0$$

$$m - \mu = 0$$

$$\boxed{m = \mu} \text{ median.}$$

vii) Show that chi-square Distribution approaches to Standard normal Distribution when degree of freedom approaches to infinity:-

Solution

If χ^2 has n degree of freedom

$$E(\chi^2) = n$$

$$V(\chi^2) = 2n$$

$$Z = \frac{\chi^2 - n}{\sqrt{2n}} \rightarrow \frac{x - \mu}{\sigma}$$

we know that m.g.f of chi-square is

$$\phi(t) = (1 - 2t)^{-n/2}$$

characteristics of Z

$$e^{tx} \cdot e^{-t(x^2 - n)/\sqrt{2n}}$$

$$E \left[e^{\frac{tx^2 - tn}{\sqrt{2n}}} \right]$$

$$e^{-tn/\sqrt{2n}} \cdot E \left[e^{tx^2/\sqrt{2n}} \right]$$



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$$= e^{-tn/\sqrt{n}} \cdot \phi_e\left(\frac{t}{\sqrt{n}}\right)$$

$$(1 - 2t)^{n/2}$$

$$= e^{-tn/\sqrt{n}} \left(1 - \frac{2t}{\sqrt{n}}\right)^{-n/2}$$

$$= \frac{-tn}{\sqrt{n}} + \left(\frac{-n}{2}\right) \ln \frac{1-2t}{\sqrt{n}}$$

$$= \frac{-tn}{\sqrt{n}} + \frac{n}{2} \left[-\ln(1-2t) \right]$$

Use series

$$= \frac{-tn}{\sqrt{n}} + \frac{n}{2} \left[\frac{2t}{\sqrt{n}} + \frac{1}{2} \left(\frac{2t}{\sqrt{n}}\right)^2 + \frac{1}{3} \left(\frac{2t}{\sqrt{n}}\right)^3 + \dots \right]$$

$$= \frac{-tn}{\sqrt{n}} + \frac{nt}{\sqrt{n}} + \frac{n^2 t^2}{2n} + \dots$$

$$= \frac{-tn}{\sqrt{n}} + \frac{nt}{\sqrt{n}} + \frac{n}{2} \left(\frac{1}{2} \right) \left(\frac{4t^2}{n} \right) + \left(\frac{n}{2} \right) \left(\frac{1}{3} \right) \left(\frac{8t^3}{n^{3/2}} \right) + \dots$$

Ignoring

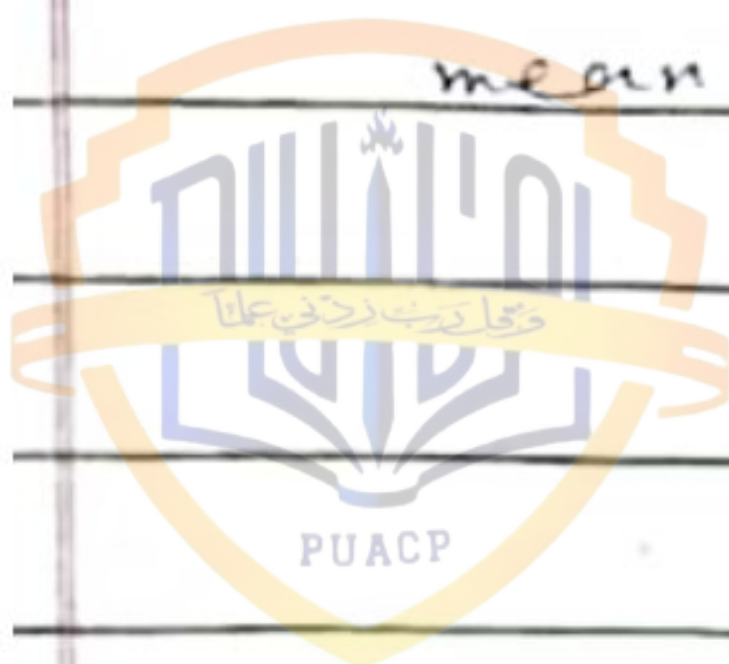
$$= t^2/2$$

$$\phi_0(t) = e^{t^2/2}$$

then moment g.f. of the
standard normal Distribution with

mean = 0 variance Hence

$$n \rightarrow \infty$$



Q-No. 03

Exponential Distribution of
m.g.f, mean, variance, γ_1, γ_2

β_1, β_2

Solution

$$M_0(t) = e^{tx}$$

$$= \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx$$
$$\lambda \int_0^{\infty} e^{-x(\lambda - t)} dx$$

$$x(\lambda - t) = z$$

$$x = \frac{z}{\lambda - t} \quad x \cdot dx = \frac{dz}{\lambda - t}$$

$$dx = \frac{dz}{\lambda - t}$$

put

$$\lambda \int_0^{\infty} e^{-z} \frac{1}{\lambda - t} dz$$

$$\lambda \cdot \frac{1}{\lambda - t} \int_0^{\infty} e^{-z} dz$$

$$\frac{\lambda}{\lambda - t} \Gamma$$

$$= \frac{\lambda}{\lambda - t}$$

$$\left(\frac{\lambda - t}{\lambda} \right)^{-1}$$

$$M_0(t) = (1 - t/\lambda)^{-1}$$



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mean

$$\begin{aligned}M_0'(t) &= (1 - t/\lambda)^{-1} \\&= -1(1 - t/\lambda)^{-2} \cdot -1/\lambda \\&= 1/\lambda (1 - t/\lambda)^{-2} \\t &= 0\end{aligned}$$

$$\text{mean} = 1/\lambda$$

variance

$$\begin{aligned}M_0(t) &= 1/\lambda (1 - t/\lambda)^{-2} \\&= -2/\lambda (1 - t/\lambda)^{-3} \cdot -1/\lambda \\&= 2/\lambda^2 (1 - t/\lambda)^{-3} \\&= 2/\lambda^2 \\2/\lambda^2 - 1/\lambda^2\end{aligned}$$

$$\text{variance} = 1/\lambda^2$$

$$M_0(t) = (1 - t/\lambda)^{-1}$$

taking ln

$$\begin{aligned}\ln M_0(t) &= -1 \ln(1 - t/\lambda) \\&= -\ln(1 - t/\lambda)\end{aligned}$$

$$= \frac{t}{\lambda} + \frac{t^2}{\lambda^2} + \frac{(t/\lambda)^3}{3} + \frac{(t/\lambda)^4}{4} + \dots$$

$$k_1 = 1/\lambda$$

$$k_2 = 1/\lambda^2$$

$$k_3 = 2/\lambda^3$$

$$k_4 = 6/\lambda^4$$

$$\gamma_1 = \frac{k_3}{(k_2)^{3/2}}$$

$$= \frac{2/\lambda^3}{(\frac{1}{\lambda^2})^{3/2}}$$

$$\boxed{\gamma_1 = 2}$$

$$\gamma_2 = \frac{k_4}{(k_2)^2}$$

$$= \frac{6/\lambda^4}{1/\lambda^4}$$

$$\boxed{\gamma_2 = 6}$$

$$\beta_1 = \lambda$$

$$u_1 = \lambda, 0$$

$$k_3 = u_3$$

$$u_2 = k_2$$

$$u_4 = k_4 + 3k_2^2$$

$$= \frac{6}{\lambda^4} + 3 \frac{1}{\lambda^2}$$

$$= \frac{9}{\lambda^4}$$

$$\beta_1 = \frac{(u_3)^2}{(u_2)^3}$$

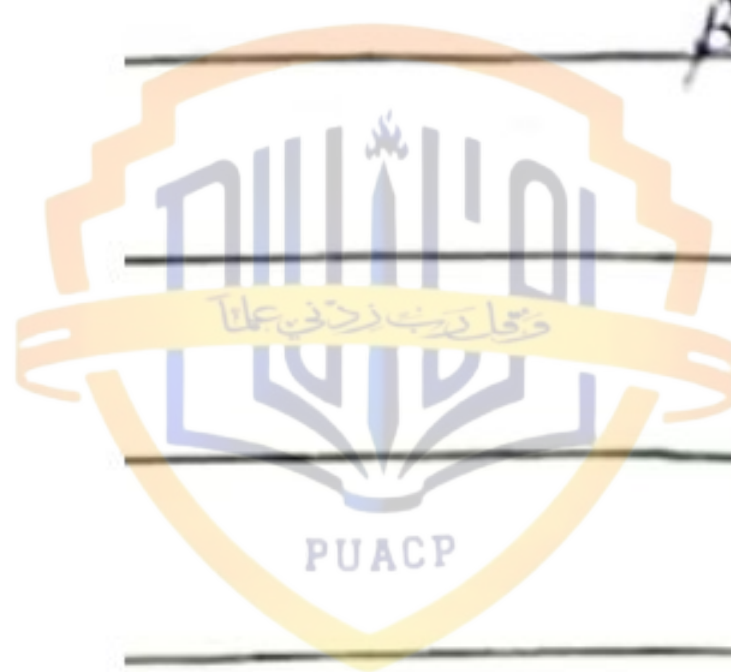
$$\frac{2/\lambda^6}{1/\lambda^6}$$

$$\boxed{\beta_1 = 2}$$

$$\beta_2 = \frac{u_4}{(u_2)^2}$$

$$= \frac{9/\lambda^4}{1/\lambda^4}$$

$$\boxed{\beta_2 = 9}$$



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Q-No-04

Probability Density Function
of F-Distribution

$$f(x_1^2) = \frac{1}{2^{\frac{n_1}{2}} \Gamma(\frac{n_1}{2})} e^{-x_1^2/2} (x_1^2)^{\frac{n_1}{2}-1}$$

$$f(x_2^2) = \frac{1}{2^{\frac{n_2}{2}} \Gamma(\frac{n_2}{2})} e^{-x_2^2/2} (x_2^2)^{\frac{n_2}{2}-1}$$

$$dF(x_1^2, x_2^2) = \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} e^{-x_1^2/2} \cdot e^{-x_2^2/2} (x_1^2)^{\frac{n_1}{2}-1} (x_2^2)^{\frac{n_2}{2}-1}$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2})} \cdot e^{-\frac{1}{2}(x_1^2+x_2^2)} (x_1^2)^{\frac{n_1}{2}-1} (x_2^2)^{\frac{n_2}{2}-1}$$

$$\frac{x_1^2/n_1}{x_2^2/n_2} = F$$

$$\frac{x_1^2}{n_1} = \frac{x_2^2}{n_2} F$$

$$x_2^2 = V \quad \frac{x_1^2}{n_1} = \frac{V}{n_2} F$$

$$\text{Hence} \quad x_1^2 = \frac{n_1}{n_2} V F$$

$$\frac{dx_1^2}{dF} = \frac{n_1}{n_2} F \quad \frac{dx_1^2}{dF} = \frac{n_1}{n_2} V$$

$$\frac{dx_2^2}{dF} = 0 \quad \frac{dx_2^2}{dF} = 1$$

PUACP

$$\left| \begin{array}{cc} \frac{n_1}{n_2} V & \frac{n_1}{n_2} F \\ 0 & 1 \end{array} \right| = \frac{n_1}{n_2} V F$$

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$$I = \left| \frac{n_1}{n} v \right|$$

put in equation

$$\frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} e^{-\frac{1}{2} \left(\frac{n_1}{n_2} v F + v \right)} \left(\frac{n_1}{n_2} v F \right)^{\frac{n_1}{2}-1} (v)^{\frac{n_2}{2}-1} \frac{n_1 v}{n_2} dv$$

$$\frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \cdot e^{-\frac{v}{2} \left(\frac{n_1}{n_2} F + 1 \right)} \cdot \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}-1} v^{\frac{n_2}{2}-1 + \frac{n_1}{2} F - \frac{n_1}{2}} dv$$

$$\frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1} \int_0^{\infty} e^{-\frac{v}{2} \left(\frac{n_1}{n_2} F + 1 \right)} v^{\frac{n_1+n_2}{2}-1} dv$$

$$\frac{v}{2} \left(\frac{n_1}{n_2} F + 1 \right) = z$$

$$v = \frac{2z}{1 + \frac{n_1}{n_2} F}$$

$$dv = \frac{2}{1 + \frac{n_1}{n_2} F} dz$$

$$\frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1} \int_0^{\infty} e^{-z} \left(\frac{2z}{1 + \frac{n_1}{n_2} F} \right)^{\frac{n_1+n_2}{2}-1} \frac{2}{1 + \frac{n_1}{n_2} F} dz$$

$$\frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1} \left(\frac{2}{1 + \frac{n_1}{n_2} F} \right)^{\frac{n_1+n_2}{2}-1} \int_0^{\infty} e^{-z} z^{\frac{n_1+n_2}{2}-1} dz$$

$$= \frac{1}{2^{\frac{n_1+n_2}{2}} \sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}}} \left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1} \left[\frac{2}{1 + \frac{n_1}{n_2} F} \right]^{\frac{n_1+n_2}{2}-1} \left[\frac{n_1+n_2}{2} \right]$$

$$\frac{\left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{\sqrt{\frac{n_1}{2}} \sqrt{\frac{n_2}{2}} \left(1 + \frac{n_1}{n_2} F \right)^{\frac{n_1+n_2}{2}}}$$

$$\frac{\left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{\beta \left(\frac{n_1}{2}, \frac{n_2}{2} \right) \left(1 + \frac{n_1}{n_2} F \right)^{\frac{n_1+n_2}{2}}}$$

$$f = \frac{\left(\frac{n_1}{n_2} \right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}}{\beta \left(\frac{n_1}{2}, \frac{n_2}{2} \right) \left(1 + \frac{n_1}{n_2} F \right)^{\frac{n_1+n_2}{2}}}$$



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$$\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}$$

$$\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) \left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}$$

$$\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} F^{\frac{n_1}{2}-1}$$

$$\frac{n_1}{2} - \frac{n_1}{2} - \frac{n_2}{2}$$

$$\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) \left(\frac{n_2 + n_1 F}{n_2}\right)^{\frac{n_1+n_2}{2}}$$

$$(n_1)^{n_1/2} (n_2)^{-n_1/2 - (n_1+n_2)/2} F^{\frac{n_1}{2}-1}$$

$$\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) (n_2 + n_1 F)^{\frac{n_1+n_2}{2}}$$

$$f(x) = \frac{(n_1)^{\frac{n_1}{2}} (n_2)^{\frac{n_2}{2}} F^{\frac{n_1}{2}-1}}{\beta\left(\frac{n_1}{2}, \frac{n_2}{2}\right) (n_2 + n_1 F)^{\frac{n_1+n_2}{2}}}$$

Bivariate marginal of X

$$= \frac{1}{\sqrt{2\pi} \sigma_1 \sigma_2 \sqrt{1-\rho^2}} \cdot \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x-\mu_1}{\sigma_1} \right) \left(\frac{y-\mu_2}{\sigma_2} \right) + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$g(x) = \frac{1}{\sqrt{2\pi} \sigma_1 \sigma_2 \sqrt{1-\rho^2}} \cdot \exp \left[-\frac{1}{2(1-\rho^2)} \left\{ \left(\frac{x-\mu_1}{\sigma_1} \right)^2 - 2\rho \left(\frac{x-\mu_1}{\sigma_1} \right) \left(\frac{y-\mu_2}{\sigma_2} \right) + \left(\frac{y-\mu_2}{\sigma_2} \right)^2 \right\} \right]$$

$$\text{let } \frac{y-\mu_2}{\sigma_2} = v$$

$$y = \mu_2 + v \sigma_2$$

$$dy = dv \sigma_2$$



Diff:

$$\frac{1}{2\pi\delta_1\sqrt{1-\beta^2}} \cdot \exp\left[-\frac{1}{2(1-\beta^2)} \left(\frac{x-u_1}{\delta_1}\right)^2 - 2\beta\left(\frac{x-u_1}{\delta_1}\right)v + v^2\right] \delta x dv$$

$$\frac{1}{2\pi\delta_1\sqrt{1-\beta^2}} \exp\left[-\frac{1}{2(1-\beta^2)} \left\{ \left(\frac{x-u_1}{\delta_1}\right)^2 + \beta^2\left(\frac{x-u_1}{\delta_1}\right)^2 - 2\beta\left(\frac{x-u_1}{\delta_1}\right)v + v^2 \right\} - \frac{\beta^2\left(\frac{x-u_1}{\delta_1}\right)^2}{2}\right] dv$$

$$\frac{1}{2\pi\delta_1\sqrt{1-\beta^2}} \cdot \exp\left[-\frac{1}{2(1-\beta^2)} \left\{ \left(\frac{x-u_1}{\delta_1}\right)^2 [1-\beta^2] + [v - \beta\left(\frac{x-u_1}{\delta_1}\right)]^2 \right\}\right]$$

$$\frac{1}{2\pi\delta_1\sqrt{1-\beta^2}} \cdot \exp\left[-\frac{\left(\frac{x-u_1}{\delta_1}\right)^2}{2} - \frac{[v - \beta\left(\frac{x-u_1}{\delta_1}\right)]^2}{2(1-\beta^2)}\right]$$

$$v - \beta\left(\frac{x-u_1}{\delta_1}\right) = w$$

$$v = \frac{w}{\sqrt{1-\beta^2}} + \beta\left(\frac{x-u_1}{\delta_1}\right)$$

$$dv = \frac{w}{\sqrt{1-\beta^2}}$$

$$\frac{1}{2\pi\delta_1\sqrt{1-\beta^2}} \exp\left[-\frac{1}{2}\left(\frac{x-u_1}{\delta_1}\right)^2 - \frac{w^2}{2}\right] \frac{w}{\sqrt{1-\beta^2}} dw$$

$$\frac{1}{2\pi\delta_1\sqrt{1-\beta^2}} \exp\left[e^{-\frac{1}{2}\left(\frac{x-u_1}{\delta_1}\right)^2} - e^{-\frac{1}{2}w^2}\right] dw$$

$$\frac{1}{\sqrt{2\pi}\delta_1} \left\{ e^{-\frac{1}{2}\left(\frac{x-u_1}{\delta_1}\right)^2} - \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}w^2} dw \right\}$$

$$g(x) = \frac{1}{\sqrt{2\pi}\delta_1} e^{-\frac{1}{2}\left(\frac{x-u_1}{\delta_1}\right)^2}$$



Date _____

Q. NO. 05 :-

If x and y are Two Independent Gamma variates with parameter (α, β_1) and (α, β_2) Then find the distribution of $z = x + y$.

Solution :-

The p.d.f of Gamma function

$$f(x) = \frac{\alpha^{\beta_1}}{\Gamma(\beta_1)} e^{-\alpha x} x^{\beta_1-1} \quad 0 < x < \infty$$

$$f(y) = \frac{\alpha^{\beta_2}}{\Gamma(\beta_2)} e^{-\alpha y} y^{\beta_2-1}$$

$$dF(x, y) = \frac{\alpha^{\beta_1} \alpha^{\beta_2}}{\Gamma(\beta_1) \Gamma(\beta_2)} e^{-\alpha(x+y)} x^{\beta_1-1} y^{\beta_2-1} dx dy$$

put $x = r \cos^2 \theta$ $y = r \sin^2 \theta$

$$x + y = r \quad z = r$$

$$\frac{dx}{dr} = \cos^2 \theta$$

$$\frac{dy}{dr} = \sin^2 \theta$$

$$\frac{dx}{d\theta} = -2r \cos \theta \sin \theta$$

$$\frac{dy}{d\theta} = 2r \sin \theta \cos \theta$$

$$|J| = 2r \cos \theta \sin \theta$$

$$= \frac{\alpha^{\beta_1} \alpha^{\beta_2}}{\Gamma(\beta_1) \Gamma(\beta_2)} e^{-\alpha(r)} (r \cos^2 \theta)^{\beta_1-1} (r \sin^2 \theta)^{\beta_2-1} 2r \cos \theta \sin \theta d\theta dr$$

$$= \frac{\alpha^{\beta_1} \alpha^{\beta_2}}{\Gamma(\beta_1) \Gamma(\beta_2)} e^{-\alpha r} r^{\beta_1-1-1-\beta_2+1} 2 \int_0^{\pi/2} (\cos \theta)^{2\beta_1-1} (\sin \theta)^{2\beta_2-1} d\theta$$

b) Five observation x_1, x_2, x_3, x_4, x_5 are drawn at random from the Distribution

$$f(x) = 20e^{-20x} \quad 0 < x < \infty$$

Find the Distribution of the following

i) the smallest ii) the largest.

Solution:-

$$f(x) = 20e^{-20x}$$

$$= \int_0^x 20e^{-20x}$$

$$= 20 \int_0^x e^{-20x}$$

$$= \left[\frac{e^{-20x}}{-20} \right]_0^x$$

$$f(x) = -e^{-20x} + e^0$$

$$F(x) = 1 - e^{-20x}$$



Date _____

$$g(x_r) = \frac{n!}{(r-1)!(n-r)!} [F(y_r)]^{r-1} f(y_r) [1-F(y_r)]^{n-r}$$

put $r=1$, $n=5$

smallest

$$\begin{aligned} &= \frac{5!}{(1-1)!(5-1)!} (20e^{-20x}) [1 - (1 - e^{-20x})]^{5-1} \\ &= 5! (20e^{-20x}) (1 - 1 + e^{-20x})^{5-1} \\ &= 5! (20e^{-20x}) (e^{-20x})^4 \\ &= 2 \times 3 \times 4 \times 5 (20e^{-20x}) (e^{-20x})^4 \\ &= 2400 (e^{-20x}) (e^{-80x}) \\ &= 2400 e^{-100x} \end{aligned}$$

$$\begin{aligned} &= n f(x_1) [1 - F(y_1)]^{n-1} \\ &= 5 (20e^{-20x}) [1 - (1 - e^{-20x})]^{5-1} \\ &= 100 e^{-20x} (e^{-80x}) \\ &= 100 e^{-100x} \end{aligned}$$

largest

$$\begin{aligned} &= n f(y_n) [F(y_n)]^{n-1} \\ &= 5 (20e^{-20x}) (1 - e^{-20x})^4 \\ &= 100 e^{-20x} (1 - e^{-20x})^4 \end{aligned}$$

